Signal Detection of Solar Maxima Using Wavelet-Enhanced Derivative Analysis in R

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Abstract:

Sunspots are the areas of the sun that contain a dark spot, and these regions are the result of intense magnetic activity known as magnetic flux, which prevents heat flow from the interior to the exterior of the Sun. During different periods of the solar cycle, the activities of sunspots vary, and this can cause a solar maximum, high activity in sunspots, or a solar minimum, low activity in sunspots. In this study, we aim to detect the daily variations leading to solar maxima from the year of 2000 to 2023. Through the use of R Studio, the first and second orders of sunspot activity will be calculated, measuring both the speed and acceleration of sunspot activity. Furthermore, these data will be used to construct a periodogram and implement a wavelet analysis to identify whether there is a presence of speed or acceleration rate that correlates with the maxima of sunspot activity. Our results show that acceleration frequencies are present in daily sunspot activity, whereas speed frequencies are not. Therefore, the detection of acceleration frequencies, rather than speed, can serve as a more reliable indicator of upcoming solar maxima. It is important to detect these acceleration frequencies as it improves forecasting of high solar activity periods and their associated risks, such as solar flares and coronal mass ejections. The significance of high solar activity is a pivotal point in major natural events, and this study aids in predicting sunspot maxima to help prevent future damage.

Background Information:

- Climate change has always been an issue worldwide, and there are many contributions to climate change, like greenhouse gases, deforestation, and industries.
- Climate change issues have been too focused on human contributions, such that many people lack the knowledge of the impacts that sunspot activities can have on climate change
- The emission of high radiation during periods of high sunspot activity can increase global warming, while low sunspot activity may cause "The Little Ice Age." (NASA, 2024)
- Sunspot activity is seen through an 11-year solar cycle along with its 5-6 periodicities (Hrushesky, W. J., Sothern, R. B., Du-Quiton, J. et al. (2011))

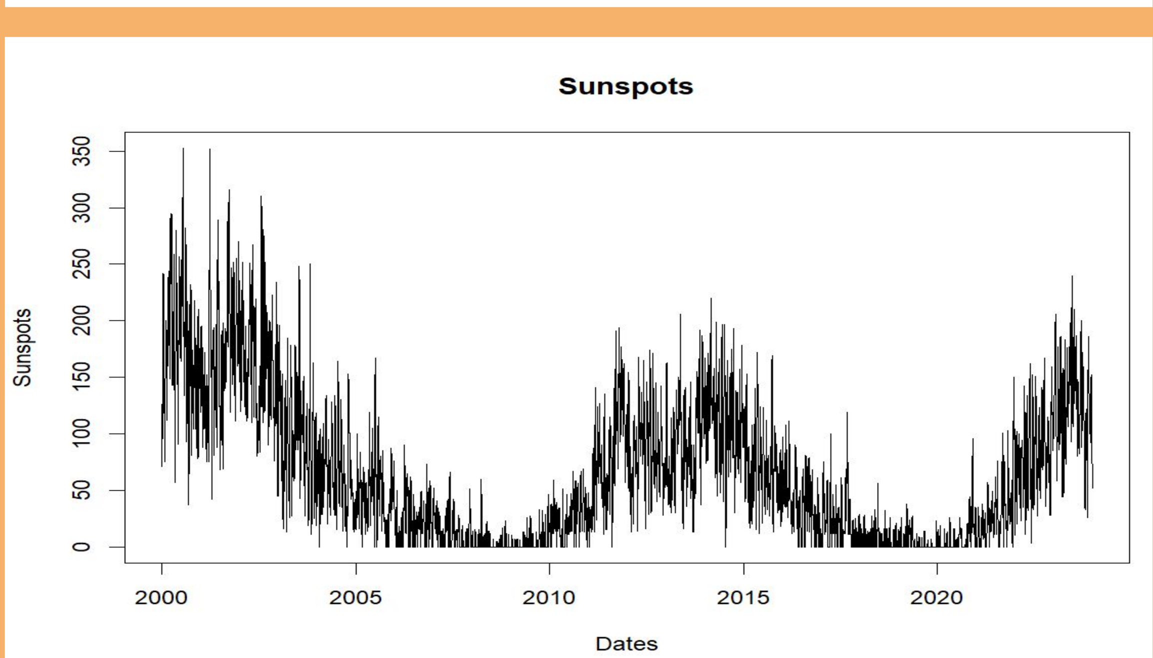

Figure 1: Sunspot activity over the past 23 years

Method/Materials:

- RStudio (v.4.5.0):
  - Utilized data from Solar Influence Data Analysis Center
  - Packages Used: data.table, TSA, WaveletComp, ggplot2, dplyr, lubridate
  - First and Second Order calculation for speed and acceleration rates
  - Spectral analysis (periodograms) and Noise Removal
  - Wavelet analysis
- Solar Influences Data Analysis Center:
  - Sunspot activity for the past 23 years in days (from 2000 to 2023)

Results:

- There is a strong frequency distinction between the acceleration rate and sunspot occurrences every 4000 days (Figure 7)
- There is a weak frequency distinction between the speed rate and sunspot occurrences every 4000 days (Figure 6)
- The frequency of max appears at 26.70623 periods for both the acceleration rate and speed rate (Figure 4 and Figure 5)

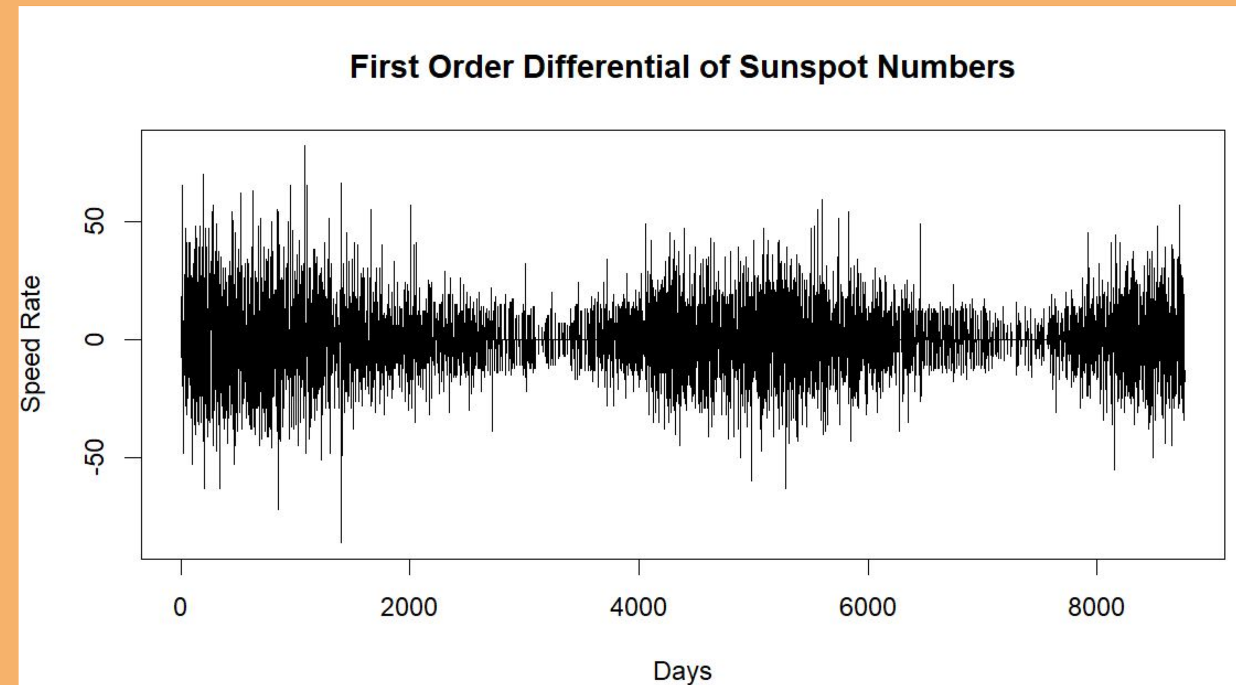

Figure 2: Daily sunspot speed rate through the first derivative calculation

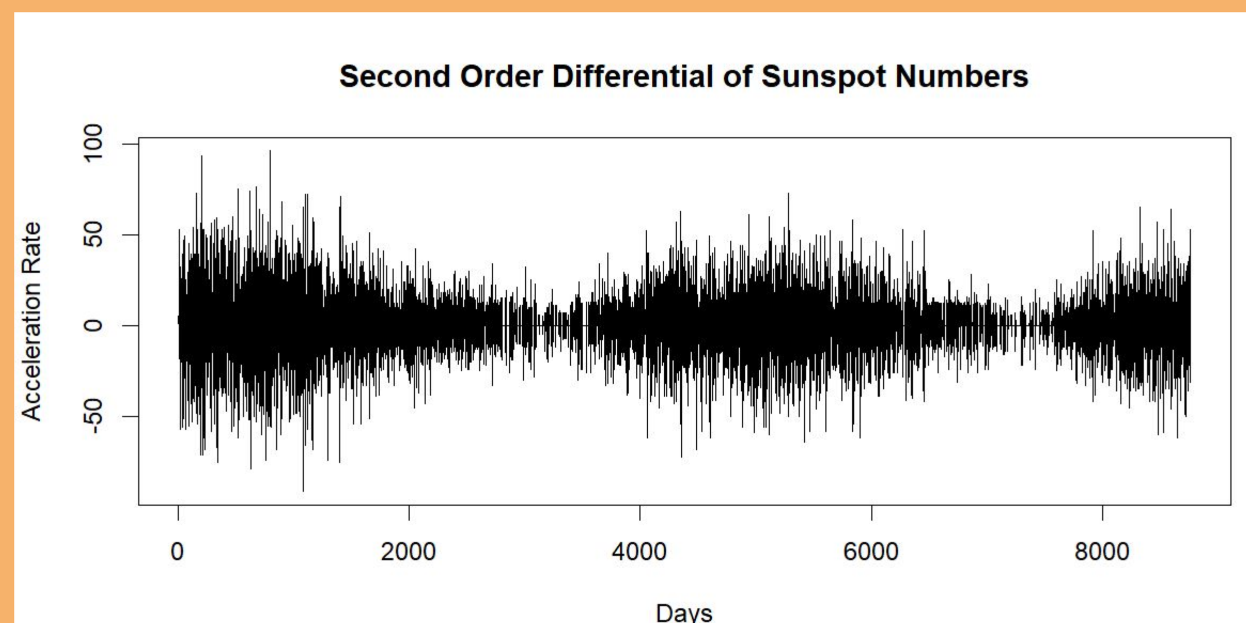

Figure 3: Daily sunspot acceleration rate through the second derivative calculation

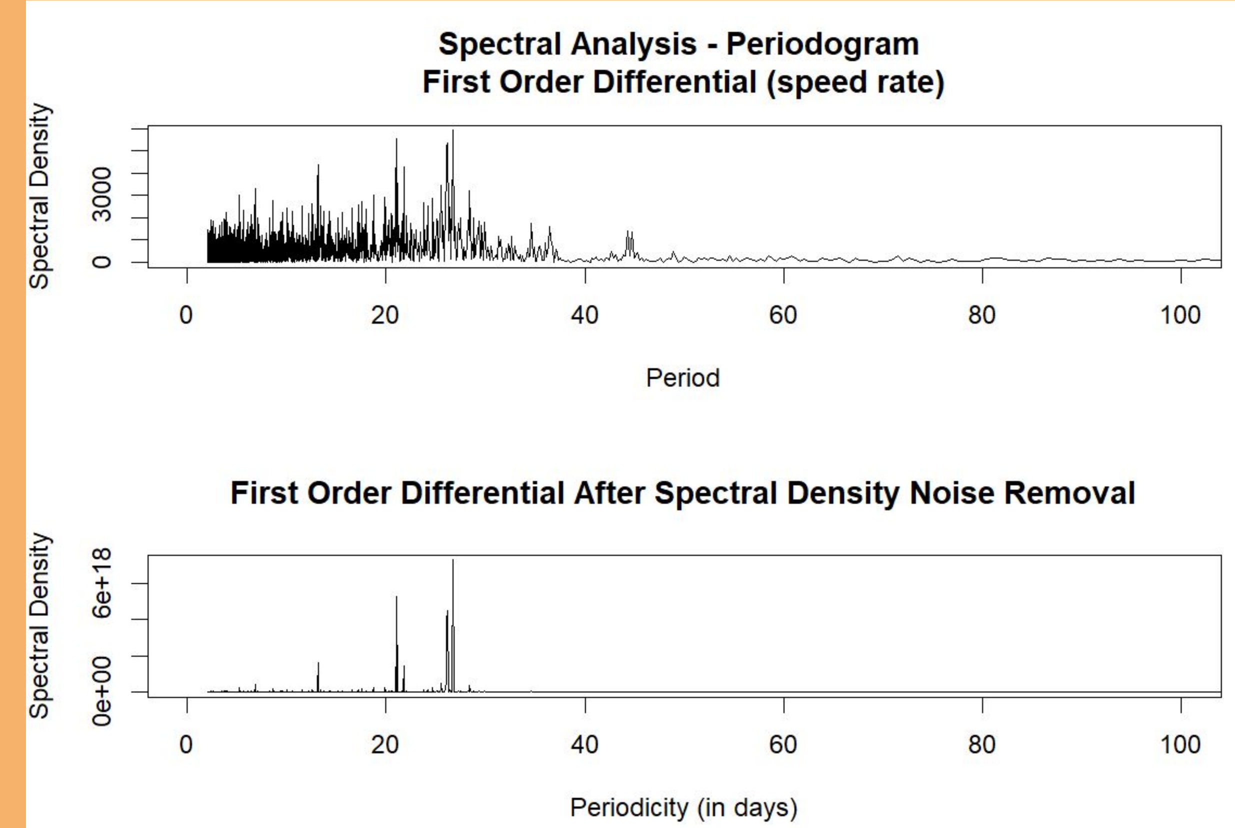

Figure 4: First-order differential frequency levels, before and after noise removal. The max frequency peak is at 337 with a periodicity of around 26.7

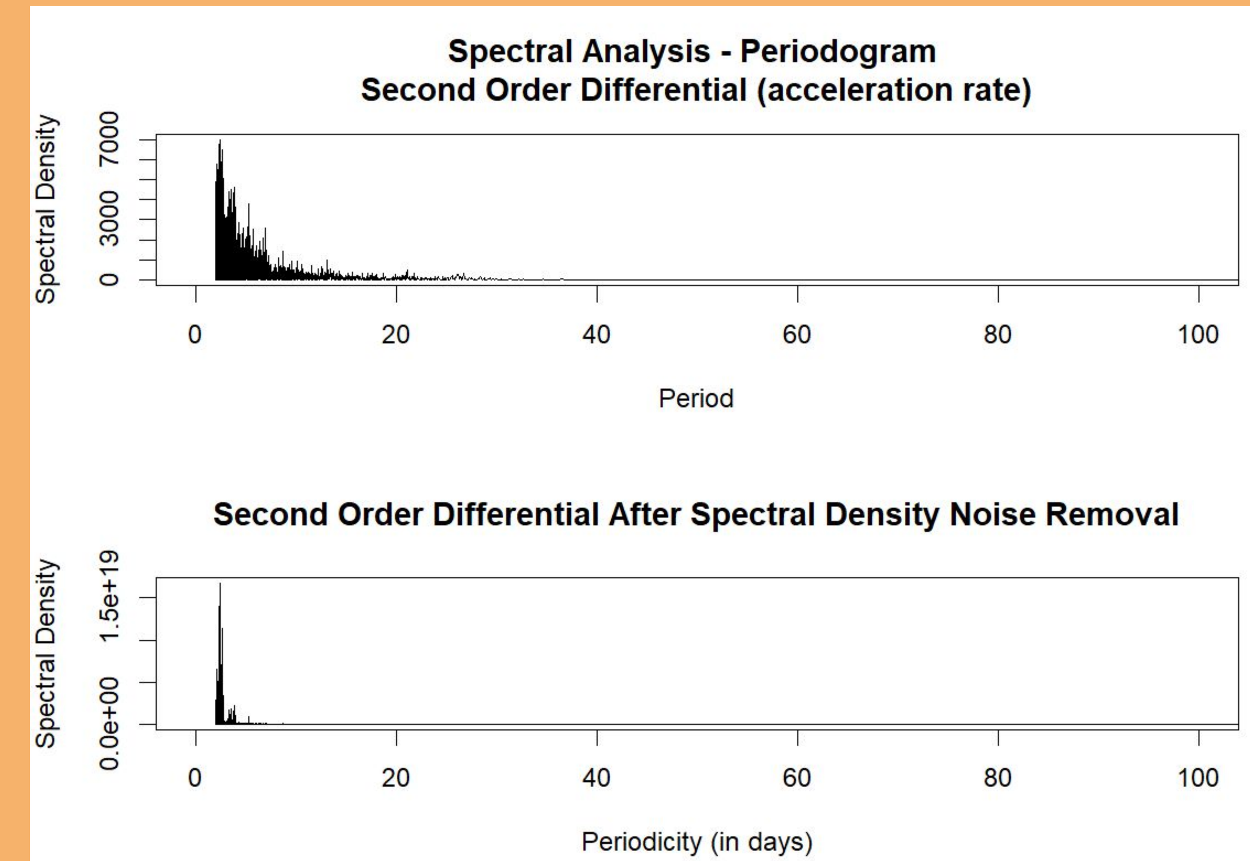

Figure 5: Second-order differential frequency levels, before and after noise removal. The max frequency peak is at 3740 with a periodicity of around 26.7

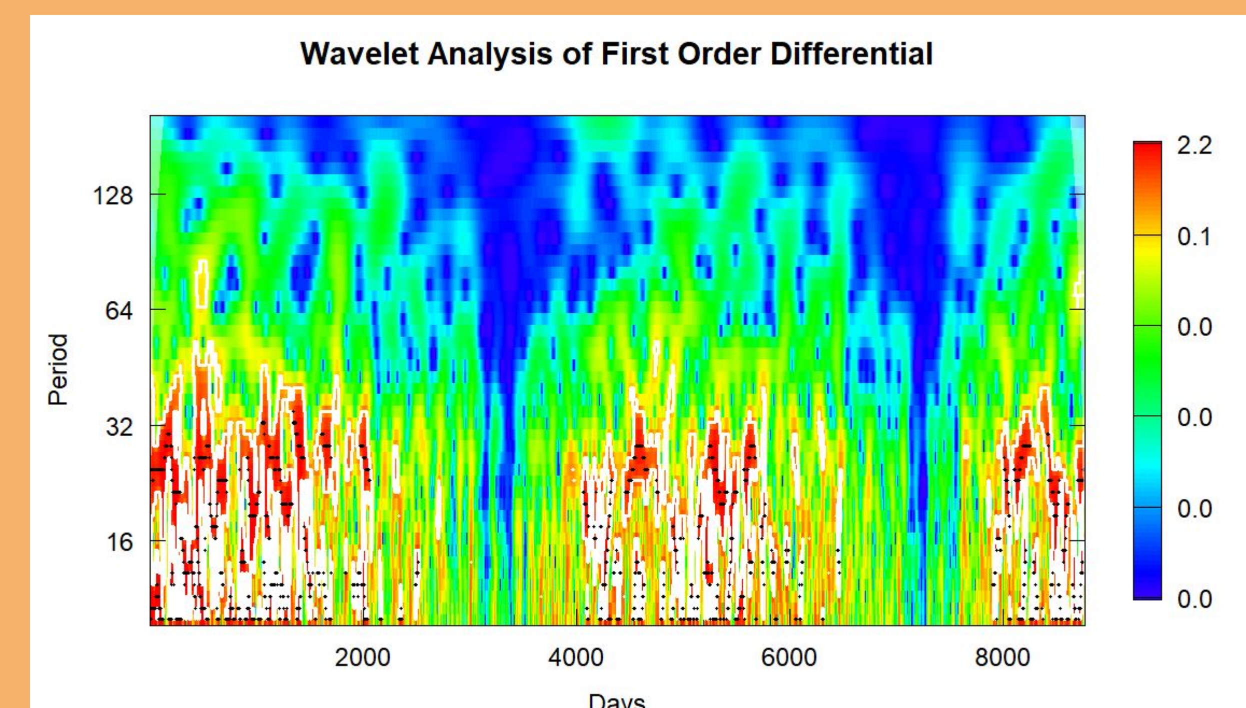

Figure 6: Wavelet analysis of the frequencies between speed rate and sunspot activity. The diagram shows a weak frequency

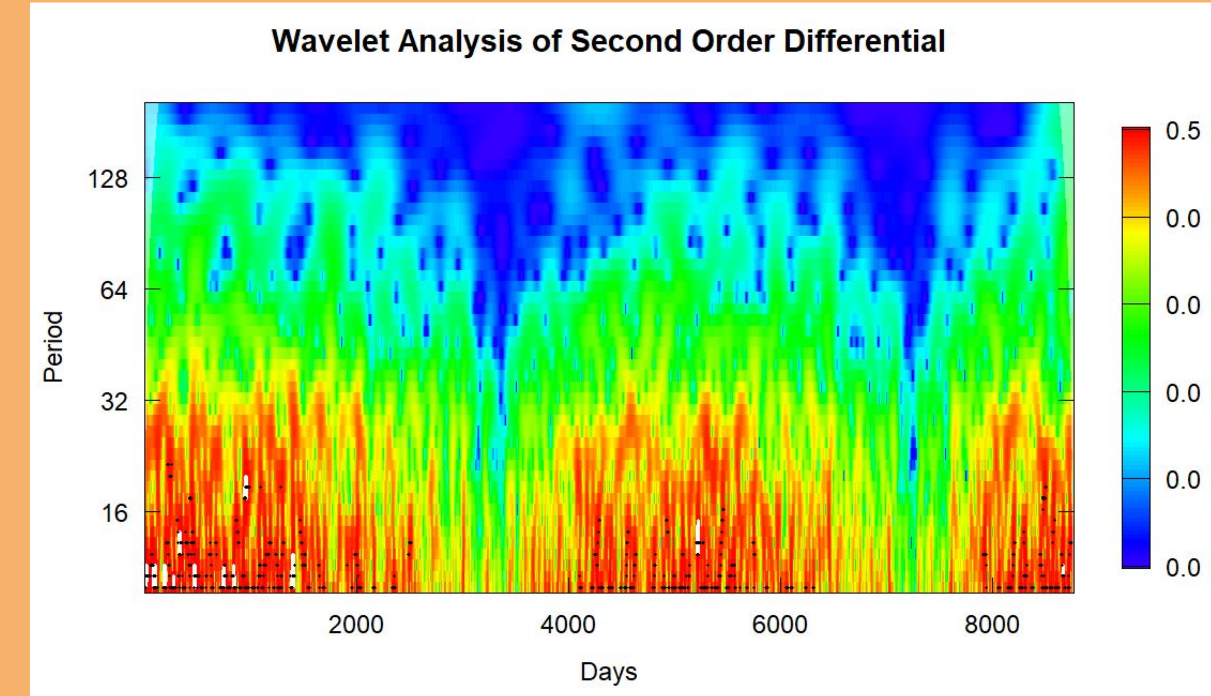

Figure 7: Wavelet analysis of the frequency between acceleration rate and sunspot activity. The diagram shows a strong frequency

Discussion/Conclusion:

- The strong frequencies shown between acceleration rate and sunspot occurrences implies that there may be a pattern present that can help detect future solar maxima
- A 4000-day (or 11-year cycle) has an impact on sunspot frequencies
- For future experiments, the use of ANN (Artificial Neural Network) can be incorporated to generate a more accurate prediction model

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Credit Authorship Contribution Statement:

This research was conducted by Ella Gao under the supervision of Prof. Marsellos. Ella Gao performed the data analysis, coding, and poster preparation, while Prof. Marsellos provided guidance, feedback, and supervision throughout the project.

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